MASTER'S THESIS
(PHYSICS)

Evolution of Open Quantum Systems under Continuous Measurement of their Environment

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- *Open quantum systems*
- *Weak measurements*
- *Stochastic master equation*
- *Quantum trajectories*
- *Photo/homodyne detection*

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« La gloire de Dieu, c'est de cacher les choses; La gloire des rois, c'est de sonder les choses. »
— **Proverbes 25:2**

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CONTEXT AND MOTIVATIONS

This document is the report of an internship carried out as part of a Master's degree in fundamental and numerical physics (CompuPhys).
My internship has been conducted within the team of Theorical Physics — *équipe de Physique Théorique* ΦTh — of the *UTINAM Institute* in the building of the *Observatory of Besançon*.
It aims to study from an academical perspective at first, the subject of continuous measurement considering open quantum systems. The main objective was to explore a subject almost-unknown in details to either my supervisor or myself. We wanted to understand the possibilities and how to handle this domain properly. This work could then lead to other searches linked to continuous measurements.

By familiarizing ourselves with the topic, we decided to adopt this perticular frame of report. The structure of this document is meant to help any unfamiliar reader (about continuous measurement), to discover and explore its fundamentals. We wanted to make it as illustrative and pedagogical as possible, thus, simple examples will be used to graphically depict mathematical and physical concepts.

Introduction

Quantum mechanics depicts the evolution of physical systems at the smallest scale in the most precise way. Indeed, this theory, built during the XXth century, allows physicists to describe and "predict" the behavior of the smallest bodies of the Universe (elementary particles: photons, electrons, etc...). Before trying to look at a complex system in a tricky situation with quantum mechanics, we need to understand what happens in the simplest case.

In the ideal case, the systems are treated as *closed*, *i.e.*, isolated from any external environment. The system is assumed to exist in a vacuum, interacting with nothing else, which enables quantum mechanics to describe its dynamics in simple terms. [1]
However in the physical world, systems do "feel" their surroundings, *i.e.*, they interact with their environment. This interaction should be taken into account to obtain a more complete and physically relevant representation. Doing so depicts the so-called *Open Quantum System* (OQS) dynamics [2]. This initial refinement can be further complemented to describe realistic physics.

Useful theories have been developed through the last decades, allowing quantum description to be more and more complete. The action of an observer interacting with a system in order to obtain information is called a measurement. Measuring a system has a tremendous impact on the quantum world. However, it is the very role of any experiment to interact with a system to gain knowledge: a measurement.
In the framework of Quantum Measurement Theory [3] one can find a whole formalism about measuring a quantum system. That is to say how the measurement acts on a system, what kind of measure the observer can read and the impact on the quantum state.

After filling these requirements, one can finally remark that measuring a quantum system alters it. The measurement usually interacts so strongly with the system that it "**collapses**" into a state that lost interesting quantum properties — an eigenstate of the observable. For example, quantum systems can be in **superposition** of states, we can think about the *Schrödinger*'s cat illustration: the cat can be alive and dead at the same time; but when one measures the state of the cat, it is either dead or alive. Then the state is "stuck" and cannot be a superposition again.
The whole motivation behind weak measurements is to measure the environment coupled with the system in order to obtain pieces of information about the system of interest. This way, information obtained should not alter too much the system due to measurement process. The measurements can even be conducted continuously, maximizing the amount of data the observer can get. These are the global ideas behind quantum continuous measurements.
We will discover throughout the following sections that two main measurements are helpful to obtain information about classical data, namely, **photodetection** and **homodyne detection** [4].

The first section will thus cover this academical progression from quantum closed systems up to the evolution of open quantum systems under continuous photodetection/homodyne detection; following the qubit and harmonic oscillator examples. The second part is a study of a system of interest on which we apply quantum continuous measurement. (Si on a le temps.)

I From closed systems to continuous measurements — in a nutshell

Test

I.i) Open Quantum Systems (OQS)

I.ii) Quantum measurement theory

I.iii) Stochastic Master Equations (SME)

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